

A Project Report
on
BIAS BUSTERS: A MACHINE LEARNING MODEL FOR
DETECTING AND MITIGATING BIAS IN NEWS ARTICLES

submitted in partial fulfillment of the requirements

for the award of degree of

BACHELOR OF TECHNOLOGY

in

Information Technology

by

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June, 2025

DECLARATION

We hereby declare that the work presented in this project entitled “**Bias Busters: A Machine Learning Model for Detecting and Mitigating Bias in News Articles**” submitted towards completion of in IV year II sem of B.Tech IT at “BVRIT HYDERABAD College of Engineering for Women”,Hyderabd is an authentic record of our original work carried out under the esteemed guidance of Dr.Aruna Rao S L, Professor & HoD , Department of Information Technology.

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CERTIFICATE

This is to certify that the Project report on “**Bias Busters: A Machine Learning Model for Detecting and Mitigating Bias in News Articles**” is a bonafide work carried out by **S. Shravani(21WH1A1250), G.Swathika (21WH1A1253), B.Harini (21WH1A1255) and D.Bhagya Laxmi(21WH1A1256)** in the partial fulfillment for the award of B.Tech degree in **Information Technology** , **BVRIT HYDERABAD College of Engineering for Women, Bachupally, Hyderabad** affiliated to Jawaharlal Nehru Technological University, Hyderabad under my guidance and supervision.

The results embodied in the project work have not been submitted to any other university or institute for the award of any degree or diploma.

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ABSTRACT

The rise of biased news content poses a threat to media integrity and public trust. This project aims to create a system that can detect bias in news articles using natural language processing , machine learning and data analytics. Our approach focuses on identifying bias indicators, such as sensationalist language, framing, and selective omission, to assess the fairness of news content. The system uses various machine learning algorithms and deep learning, to classify articles based on their bias levels, helping to reveal the elements contributing to the bias. We conduct experiments to test how well the system performs, with a focus on accuracy and the ability to recognize different types of bias. Early results show that our method outperforms current bias detection models by effectively identifying subtle biases. This work aims to promote fair media practices and provide a valuable tool for journalists, researchers, and readers.

Keywords: NLP,ML,BERT,LSTM

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LIST OF ABBREVIATIONS

Abbreviation	Acronym
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
BERT	Bidirectional Encoder Representations from Transformers
T5	Text-to-Text Transfer Transformer
SVM	Support Vector Machine
UI	User Interface
API	Application Programming Interface
CSV	Comma Separated Values
RAM	Random Access Memory
IDE	Integrated Development Environment
OS	Operating System
KMeans	K-Means Clustering Algorithm
POS	Part of Speech
NLTK	Natural Language Toolkit

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1. INTRODUCTION

In an era where digital media has become the cornerstone of global communication, the spread of information has reached unprecedented levels of speed, scale, and accessibility. However, this transformation has also brought with it a significant challenge: the rise of biased, misleading, and manipulative content. In particular, news articles—traditionally regarded as pillars of truth and objectivity—are increasingly subject to various forms of bias that can distort public perception and influence social behavior. The infiltration of bias into journalism poses a serious threat to democratic discourse, critical thinking, and the ability of citizens to make informed decisions.

Bias in news can appear in multiple guises. These include overt forms such as politically skewed narratives, emotionally charged language, and selective quotation, as well as subtler mechanisms like framing effects, topic omission, and unequal representation of perspectives. These biases often reflect the ideologies of media owners, political affiliations, or commercial interests. As such, bias is not always intentional—it can also be an unconscious outcome of editorial practices or systemic cultural patterns. Regardless of its source, biased reporting erodes the credibility of journalism and contributes to the spread of misinformation and social polarization.

With the proliferation of social media and algorithmically driven news feeds, biased content has found fertile ground to thrive. Recommendation systems tend to prioritize engagement over accuracy, thus amplifying sensational or emotionally provocative content. Readers are frequently exposed to echo chambers where their existing beliefs are reinforced while dissenting views are filtered out. This echo effect weakens societal cohesion and hinders open, evidence-based discussion. In such an environment, it becomes imperative to develop technological interventions that help identify and reduce bias in media content.

Traditional methods of bias detection, such as manual content analysis by experts, are time-consuming, costly, and difficult to scale. They also suffer from subjective interpretations, as even trained analysts

may disagree on what constitutes bias. To overcome these limitations, recent research has turned to artificial intelligence (AI), particularly natural language processing (NLP), to automate the task of bias detection. By leveraging machine learning algorithms, it is now possible to process vast quantities of news data and detect linguistic patterns that signal the presence of bias with high accuracy and consistency.

This project proposes a comprehensive framework that harnesses the power of machine learning and NLP to detect, classify, and mitigate bias in news articles. The system processes raw text inputs and classifies them as biased or unbiased using a combination of classical models (such as TF-IDF and logistic regression) and transformer-based deep learning models (such as BERT). Once bias is detected, the system provides mitigation strategies either by flagging problematic language or suggesting reworded, neutral alternatives using generative models like T5. In addition to text analysis, the project also includes a visual component that evaluates image-based media for demographic representation, thus extending bias detection to multimodal formats.

The primary goal of the system is to enable journalists, researchers, policy-makers, and everyday readers to assess the neutrality of the content they consume and create. It empowers users to become more critical thinkers and conscientious communicators by providing transparent explanations for bias classification and actionable suggestions for improvement. This is particularly important in sensitive domains such as political reporting, health communication, and social justice, where biased narratives can have far-reaching implications.

Beyond technical implementation, the project aspires to contribute to broader societal goals. Media serves as a mirror to society, but it also shapes how people perceive the world. Unchecked bias can perpetuate stereotypes, marginalize communities, and escalate conflicts. By building a tool that promotes fair and balanced reporting, this initiative aligns with the ethical imperative of journalism: to inform without distortion. It also supports educational efforts by helping students, educators, and

researchers understand how language choices can subtly frame narratives and influence interpretation.

The project methodology includes extensive data preprocessing, supervised learning for classification, unsupervised clustering for pattern discovery, and advanced text generation for rewriting. It utilizes labeled datasets of news articles, custom bias term libraries, and pretrained models for contextual understanding. The system is deployed via a user-friendly web interface using Gradio, making it accessible to non-technical users.

In summary, this project addresses a pressing need in modern information ecosystems: the ability to objectively analyze and reduce bias in news content. It combines technical innovation with social relevance, drawing from computational linguistics, ethical AI, and media studies. By promoting transparency, fairness, and accountability in news reporting, the project aims to strengthen public trust in journalism and uphold the foundational principles of democratic dialogue.

1.1. Objective

This project seeks to harness machine learning and natural language processing to develop an automated system for detecting and mitigating bias in news articles. Using advanced machine learning models, the system will classify news content based on bias levels and identify the key linguistic features contributing to the bias. The primary objectives include:

1. Developing a robust model capable of identifying various types of bias, such as sensationalism, framing, and omission.
2. Ensuring high accuracy and reliability in detecting subtle biases within diverse datasets.
3. Providing a user-friendly interface for journalists, researchers, and readers to evaluate the fairness of news content.

1.2. Problem Definition

The increasing prevalence of biased news content poses challenges to maintaining media integrity and public trust. Conventional methods for detecting bias are subjective, inconsistent, and resource-intensive, making them unsuitable for large-scale or real-time applications. These constraints highlight the need for an automated, scalable, and objective system capable of identifying and mitigating bias in news articles. This project focuses on developing a machine learning-based model to detect and classify bias in news content. Leveraging Natural Language Processing techniques and state-of-the-art architectures, such as BERT and RoBERTa, the proposed system aims to identify key indicators of bias, including sensationalist language, selective omission, and framing, provide high accuracy in classifying articles based on bias levels, and enable real-time analysis and scalability for diverse datasets. The system aims to foster a more transparent and trustworthy information ecosystem, ensuring that media content aligns with the principles of fairness and integrity.

1.3. Modules

1. Data Preprocessing
2. Feature extraction
3. Bias Classification
4. Bias Detection and Interpretation
5. Bias Mitigation
6. Model Evaluation
7. User Interface
8. Image Bias Detection

2. LITERATURE SURVEY

Dbias: Detecting Biases and Ensuring Fairness in News Articles ^[1] presents Dbias, a machine learning tool designed to identify and reduce bias in news content. Dbias works through multiple steps, starting with detecting biased language in text. It then pinpoints specific biased words, masks them, and replaces them with neutral alternatives to create more balanced content. The tool ensures fairness at every stage, including data preparation, model training, and final analysis. The authors emphasize that machine learning models can unintentionally learn and amplify biases present in large datasets, which might not be easily noticed by developers. Dbias solves this issue by offering a built-in system to detect and fix biases, ensuring that fairness is maintained throughout the entire process. This approach is particularly important for creating more objective and reliable news articles, as biased content can influence public perception. By integrating algorithms for bias detection and correction, Dbias helps make machine learning workflows more ethical and transparent, addressing a critical challenge in the modern use of Artificial Intelligence for news and media. The paper highlights the importance of tools like Dbias to promote fairness and accountability in automated systems.

The Effects of Media Bias in Recommendation Systems ^[2] investigates the interplay between media bias and personalized news recommendation algorithms. The research focuses on state-of-the-art neural network models like NAML, NRMS, and PLM-empowered algorithms, which use user history and metadata to generate recommendations. The study reveals how these algorithms often amplify biased content when catering to user preferences, potentially creating echo chambers and filter bubbles. A key finding is that the volume of biased news recommended to users is influenced by their historical consumption of biased media. The study evaluates recommendation performance using metrics such as Normalized Discounted Cumulative Gain (NDCG) and Mean Reciprocal Rank (MRR), while also analyzing the algorithms' sensitivity to user preferences. The authors suggest integrating fairness-aware mechanisms into recommendation systems to counteract the propagation of media bias, thereby promoting balanced and trustworthy news consumption.

Geo-Political Bias in Fake News Detection AI: The Case of Affect^[3] explores both human and machine abilities to detect fake news. The research compares the performance of humans and machine learning models when provided with different types of article information, such as titles, images, source biases, and text excerpts. It finds that humans are better at identifying fake news when aided by metadata like images or source information, while machine learning models outperform humans across all scenarios. Key factors affecting human judgment include cognitive biases, confirmation bias, and attentional limitations. The study employs the FakeNewsNet dataset, annotated by Politifact and BuzzFeed, to train machine learning models. It also emphasizes the need for platforms to incorporate indicators and warnings to help users make informed decisions. Additionally, the research highlights the potential for integrating metadata-based approaches into automated fake news detection systems to improve their effectiveness and reach.

Bias in Artificial Intelligence: Sources, Challenges, and Mitigation Strategies ^[4] provides a comprehensive examination of bias in AI systems, particularly emphasizing socio-political contexts where biased decisions can have significant societal impacts. The paper categorizes biases into historical, representation, and measurement bias, describing how these emerge from skewed data, flawed algorithms, and systemic inequalities. It discusses how bias permeates all AI development stages—from data collection, feature selection, to model deployment—and how opaque “black-box” models can exacerbate bias issues due to lack of transparency. The authors propose mitigation strategies such as preprocessing techniques to rebalance data distributions, fairness-aware in-processing algorithms, and post-processing adjustments to model outputs. Ethical and regulatory challenges are highlighted, stressing the need for collaboration among technologists, ethicists, and policymakers to ensure responsible AI deployment. The paper also underscores the importance of explainable AI (XAI) and regular auditing of AI systems to detect and correct biases proactively. It concludes with actionable recommendations to embed fairness-by-design principles in AI workflows to foster equitable and trustworthy AI solutions.

Towards Better Detection of Biased Language with Scarce, Noisy Annotations [5] addresses the challenge of detecting biased language in situations where annotated datasets are limited and noisy, which is common in real-world applications. The paper introduces CLEARE, a novel framework that combines contrastive learning with data augmentation to enhance biased language detection accuracy. CLEARE leverages contrastive learning to better distinguish subtle biases in language by learning robust representations even with scarce and imperfect labels. The authors detail the difficulties of capturing nuanced biased expressions, particularly in social media and news text, where biased language may be implicit. Experiments conducted on benchmark datasets demonstrate that CLEARE outperforms previous methods, achieving higher detection accuracy and improved generalization. The framework's robustness to noisy annotations suggests it is practical for real-world deployment where expert labeling is expensive. The study emphasizes the importance of such methods to develop fairer natural language processing tools capable of reducing the propagation of harmful stereotypes and prejudice in automated systems.

Bias-aware Face Mask Detection Dataset [6] focuses on the demographic biases present in face mask detection models that emerged during the COVID-19 pandemic. The authors introduce the Bias-Aware Face Mask Detection (BAFMD) dataset, which includes 6,264 images with 13,492 masked faces and 3,118 unmasked faces collected from Twitter worldwide. Unlike previous datasets biased towards specific racial or ethnic groups, BAFMD offers balanced representation across race, gender, and age. This balance is critical because prior models showed decreased accuracy for underrepresented groups, raising fairness concerns. The dataset includes varied real-world conditions, such as lighting and pose variations, enhancing model robustness. By applying state-of-the-art face attribute prediction techniques, the authors ensure demographic diversity and quality control. Experimental results validate that models trained on BAFMD outperform those trained on traditional datasets, achieving fairer and more accurate detection. This dataset represents a significant contribution toward equitable computer vision systems, especially important for public health applications during pandemics.

VERITE: A Robust Benchmark for Multimodal Misinformation Detection Accounting for Unimodal Bias ^[7] introduces VERITE, a comprehensive benchmark for misinformation detection that addresses the challenge of unimodal bias present in many existing multimodal datasets. The authors observe that prior datasets often allow models to rely disproportionately on one modality (text or image), resulting in biased or unreliable detection performance. VERITE proposes synthetic data augmentation and novel evaluation protocols designed to reduce unimodal bias, ensuring that detection models genuinely learn cross-modal interactions. The benchmark includes diverse misinformation types and is crafted to evaluate model robustness in real-world scenarios where multimodal signals may conflict. The paper also explores various state-of-the-art multimodal architectures and highlights their strengths and limitations on VERITE. By providing this dataset and benchmark, the authors contribute significantly to advancing trustworthy and effective misinformation detection methods that do not rely on shortcuts or modality biases.

3. SYSTEM DESIGN

We train and evaluate our architecture using a labeled dataset of news articles that contains both biased and non-biased samples. The dataset includes raw textual content along with corresponding labels that indicate whether the article demonstrates any form of bias. Each news article was preprocessed to remove noise, such as punctuation, URLs, special characters, and stopwords. The cleaned data was then lemmatized using spaCy and converted to lowercase for standardization.

The architecture uses two types of feature extraction methods: traditional TF-IDF vectorization and contextual embeddings from pre-trained language models like BERT and RoBERTa. The initial classification is performed using a logistic regression model trained on TF-IDF features. For improved accuracy, we later fine-tune a BERT-based model on the cleaned and tokenized dataset.

To interpret and classify bias, we identify biased terms and phrases in each article and categorize them based on the type of bias — such as gender, racial, socioeconomic, age, or disability bias. A predefined list of biased terms and phrases is matched against article content. Additionally, the model uses attention mechanisms and prediction weights to infer the level of bias, categorized as low, medium, or high.

For mitigation, we apply both rule-based and model-based techniques. The rule-based module uses predefined replacements for biased terms, while the model-based method utilizes a fine-tuned T5 model to rewrite biased content into a more neutral and inclusive form. This helps retain the core message while eliminating potentially discriminatory language.

Finally, a user interface is developed using Gradio that allows users to input text and view detailed feedback. The UI highlights detected biased terms, provides a severity level, and suggests neutral rewrites. For transparency and interpretability, evaluation metrics such as accuracy, precision, recall, and F1-score are calculated using confusion matrices and classification reports.

The system thus provides an end-to-end pipeline for detecting, explaining, and mitigating bias in news content using both classical machine learning and modern transformer-based architectures.

3.1 Architecture

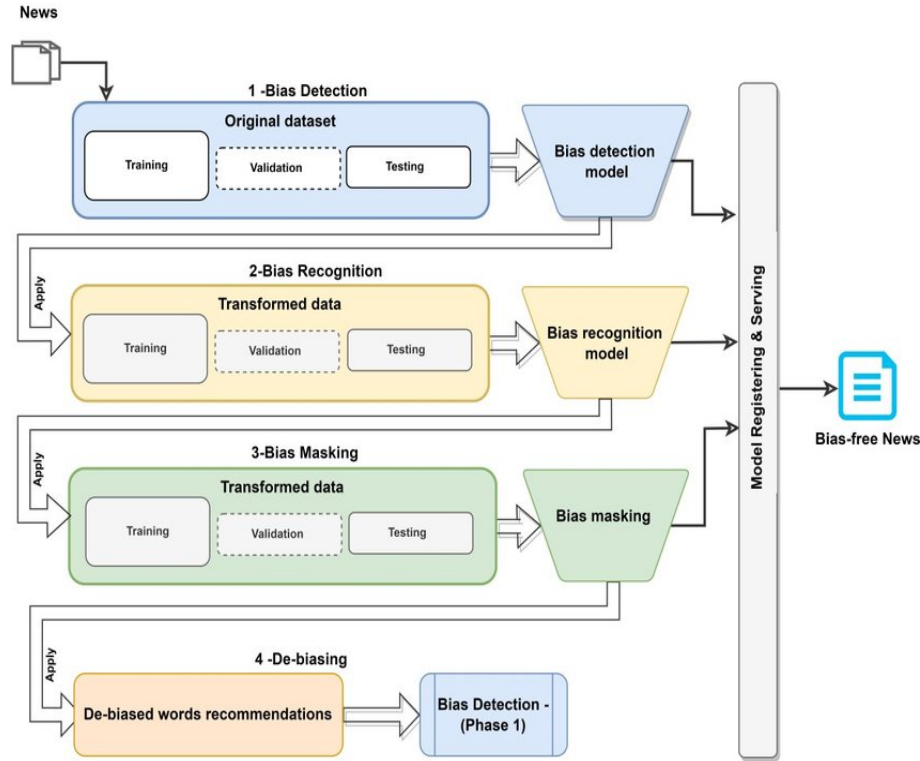


Figure 3.1: System Architecture for Bias Detection and Mitigation

The system architecture for bias detection and mitigation is organized into four key phases: Bias Detection, Bias Recognition, Bias Masking, and De-biasing. These phases work in sequence to process raw news articles and produce bias-free rewritten outputs. The architecture utilizes state-of-the-art transformer models such as BERT, RoBERTa, and T5, supported by traditional NLP techniques

and modular APIs for model serving.

Bias Detection Phase

In the first phase, original news articles are preprocessed using standard text cleaning techniques including lowercasing, removal of special characters, stopword elimination, and lemmatization. The cleaned text is used to train a transformer-based classification model (e.g., BERT) or a traditional model such as logistic regression with TF-IDF features. The model categorizes each article as either biased or non-biased. This serves as the entry point to the bias mitigation pipeline.

Bias Recognition Phase

The second phase focuses on recognizing specific biased terms within the identified biased content. This is achieved using a transformer model trained to tag and highlight words or phrases that contribute to bias, such as those related to gender, race, or political leaning. Predefined keyword lists are also used for cross-verification. The output of this phase is a set of recognized biased words within the input article.

Bias Masking Phase

In this phase, the recognized biased terms are masked using another transformer model trained specifically for this purpose. The goal is to blank out or substitute the biased words while retaining the overall structure of the sentence. This masking prepares the sentence for neutral rephrasing in the next phase.

De-biasing Phase

The final phase involves generating a neutral version of the masked sentence using a T5-based sequence-to-sequence model. This model fills in the masked sections with unbiased and contextually accurate alternatives. The result is a rewritten, de-biased news article that maintains the intended message while promoting fairness and inclusiveness. Each phase is deployed as a separate inference

API: Bias Detection API, Bias Recognition API, and Bias Masking API. These APIs are modular and support integration into real-time systems or research platforms. The overall architecture is scalable, supports transfer learning, and enables various downstream applications.

3.1.1 Bias Detection in Text Using Machine Learning and Transformers

This section describes the implementation of a bias detection system using a combination of machine learning techniques and transformer-based language models. The process involves data preprocessing, vectorization, clustering, classification, and bias mitigation.

The dataset, consisting of text samples, was first cleaned using regular expressions and Natural Language Toolkit (NLTK) to remove stop words, punctuation, and URLs. The cleaned text was then vectorized using TF-IDF (Term Frequency–Inverse Document Frequency) and clustered using the K-Means algorithm to identify biased versus unbiased content.

To build a robust classifier, a logistic regression model was trained and evaluated using metrics such as accuracy and classification reports. The final bias classification task was also implemented using a fine-tuned BERT model via Hugging Face Transformers. Tokenization was applied with ‘AutoTokenizer’, and a sequence classification model was trained using ‘Trainer’ and ‘TrainingArguments’.

Additionally, bias mitigation was explored using both rule-based replacement of biased terms and neural text generation. The T5 model (‘t5-small’) was employed to rewrite biased sentences in a neutral tone.

To evaluate the performance, confusion matrices and classification reports were generated. Furthermore, an interactive Gradio-based UI was developed for both text and image bias detection,

enabling real-time analysis and mitigation with visualization of bias level, terms, and categories.

The model also included an image bias detection module using simulated demographic distribution analysis to flag underrepresentation of gender and ethnic groups, enhancing the inclusivity of multimedia content.

3.2 Technologies

3.2.1 Natural Language Processing (NLP)

Natural Language Processing (NLP) is a crucial domain within artificial intelligence that focuses on enabling machines to understand, interpret, and generate human language. In this project, NLP techniques were essential for preprocessing text data. This involved cleaning the raw text by removing punctuation, stopwords, and irrelevant symbols. The ‘spaCy’ library was employed to perform tokenization, part-of-speech tagging, and linguistic feature extraction. These preprocessing steps helped enhance the quality of input data, ensuring more accurate bias detection by the subsequent machine learning and deep learning models.

3.2.2 Scikit-learn

Scikit-learn, a robust and widely used Python library for machine learning, played a central role in the initial stages of model building. The project utilized scikit-learn for transforming the cleaned text into numerical vectors using the TF-IDF (Term Frequency–Inverse Document Frequency) technique. These vectors served as features for the machine learning model. The KMeans clustering algorithm was then applied to group text samples into biased and unbiased categories. Furthermore, a logistic regression classifier was trained for supervised bias classification. Model performance was evaluated through metrics such as accuracy, confusion matrix, and classification report, all implemented using scikit-learn’s built-in functions.

3.2.3 Transformers (Hugging Face)

The Hugging Face ‘transformers’ library enabled the implementation of state-of-the-art transformer-

based models in the project. Two pre-trained models were utilized—BERT and T5. BERT (Bidirectional Encoder Representations from Transformers) was fine-tuned to perform sequence classification, identifying whether a given text contains biased content. Meanwhile, T5 (‘t5-small’) was used for bias mitigation through text generation, which involved rephrasing biased sentences into more neutral and inclusive forms. The ‘AutoTokenizer’ class was employed to tokenize inputs according to the respective models’ requirements, ensuring seamless compatibility and effective processing.

3.2.4 PyTorch

PyTorch served as the deep learning framework for training and evaluating transformer models in the project. Custom dataset classes were defined using PyTorch to handle the tokenized data and associated labels. These classes integrated smoothly with the Hugging Face ‘Trainer’ API, enabling efficient training workflows. During evaluation and inference, PyTorch handled the input feeding, model execution, and output processing. The use of PyTorch facilitated flexible model development and made it possible to fine-tune large language models effectively for bias classification.

3.2.5 Gradio

To enhance the usability and accessibility of the bias detection system, Gradio was employed to build a graphical user interface. This interface enabled users to input text and receive real-time bias analysis, including highlighted biased terms and suggested rewritten alternatives. Additionally, Gradio supported image upload functionality to analyze visual content for representation bias based on gender and ethnicity. The interactive nature of the Gradio interface made it suitable for both demonstration and practical deployment.

3.2.6 Bias Categories and Mitigation

The project addressed various forms of bias including racial, gender, socioeconomic, disability, and age-related biases. Each category was associated with a curated list of biased terms known to reflect harmful stereotypes or discrimination. To mitigate these biases, the project employed a two-pronged approach. First, a rule-based system replaced identified biased terms with more neutral alternatives. Second, for more complex or context-dependent biases, the T5 transformer model was used to automatically rewrite entire sentences to make them bias-free while retaining their original meaning.

3.2.7 Confusion Matrix and Evaluation

To ensure the effectiveness of the bias classification models, comprehensive evaluation metrics were employed. A confusion matrix was generated using the Seaborn library to visualize the model's performance in distinguishing between biased and unbiased content. Additional metrics such as precision, recall, F1-score, and accuracy were calculated using scikit-learn to provide a detailed assessment of the classifier's reliability. These evaluations confirmed that the model was capable of making consistent and accurate predictions.

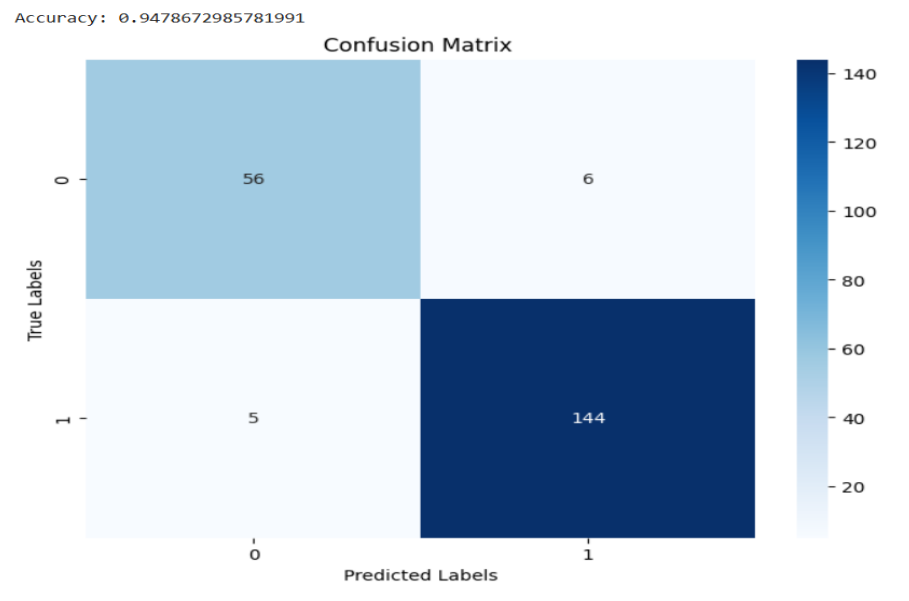


Figure 3.2: Confusion Matrix for Bias Detection Classifier

3.3 UML Diagrams

To understand the structure and functionality of the bias detection and mitigation system, we describe its behavior using Unified Modeling Language (UML) diagrams. These diagrams help visualize user

interactions, system workflows, and the decision logic involved in processing and mitigating bias in news content.

3.3.1 Use Case Diagram

The Use Case diagram illustrates how different users interact with the system. The primary actors are the *End User*, *Data Scientist*, and *System Administrator*. The End User can submit text or images to detect and mitigate bias. The Data Scientist can retrain models and evaluate system performance. The System Administrator is responsible for deploying and maintaining the system. Optional use cases such as adjusting confidence thresholds and language settings are also included.

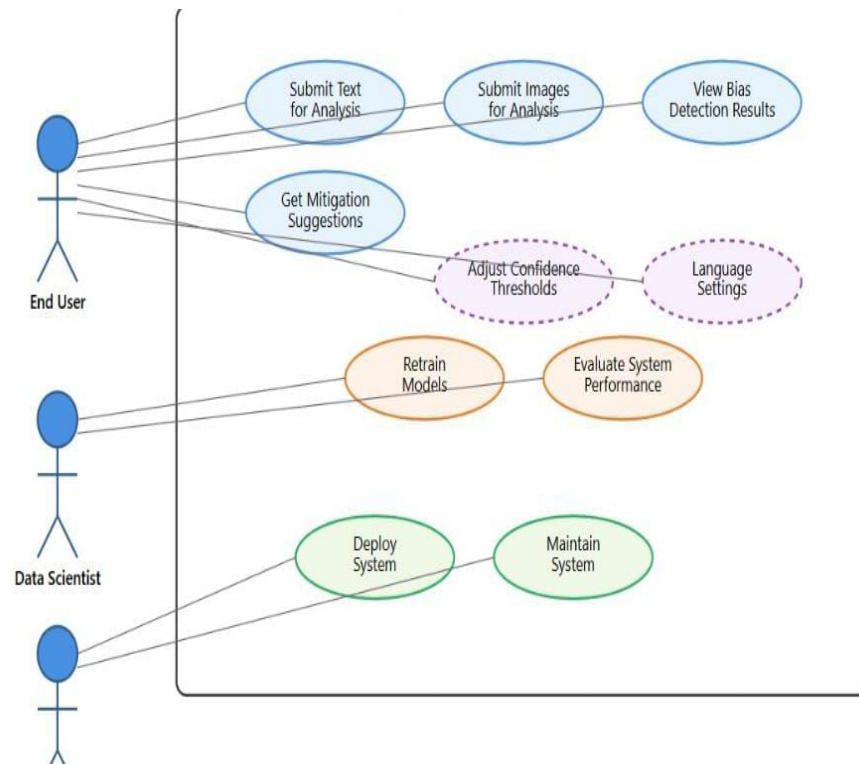


Figure 3.3: Use Case Diagram

3.3.2 Sequence Diagram – Text Bias Analysis

The Sequence diagram describes the flow of actions during a text bias analysis request. The user inputs

a text snippet through the Gradio interface. The input is passed to the text preprocessing module, which cleans and tokenizes the text. The cleaned data is then sent to both a traditional classifier using TF-IDF and a deep learning-based BERT classifier. Predictions from both are consolidated by a bias aggregation module. If bias is detected, the mitigation module (either rule-based or T5 model) generates a neutral version, which is then returned to the user.

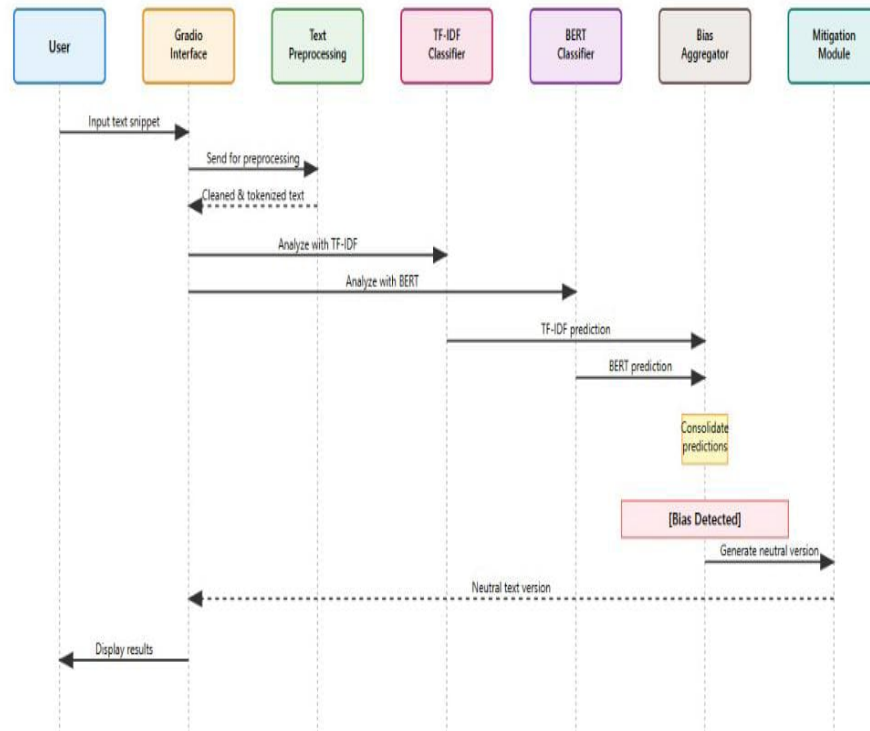


Figure 3.4: Sequence Diagram

3.3.3 Activity Diagram – Bias Mitigation Work Flow

The Activity diagram presents the logic flow for mitigating bias once it has been identified. The workflow splits into two paths: a rule-based replacement of biased words using a predefined dictionary, and a model-based paraphrasing path that utilizes a T5 transformer to rewrite the sentence. Both paths converge at a validation checkpoint to ensure that the final result is neutral. If bias remains,

the process loops through stricter rules or refined model parameters until the output meets neutrality standards.

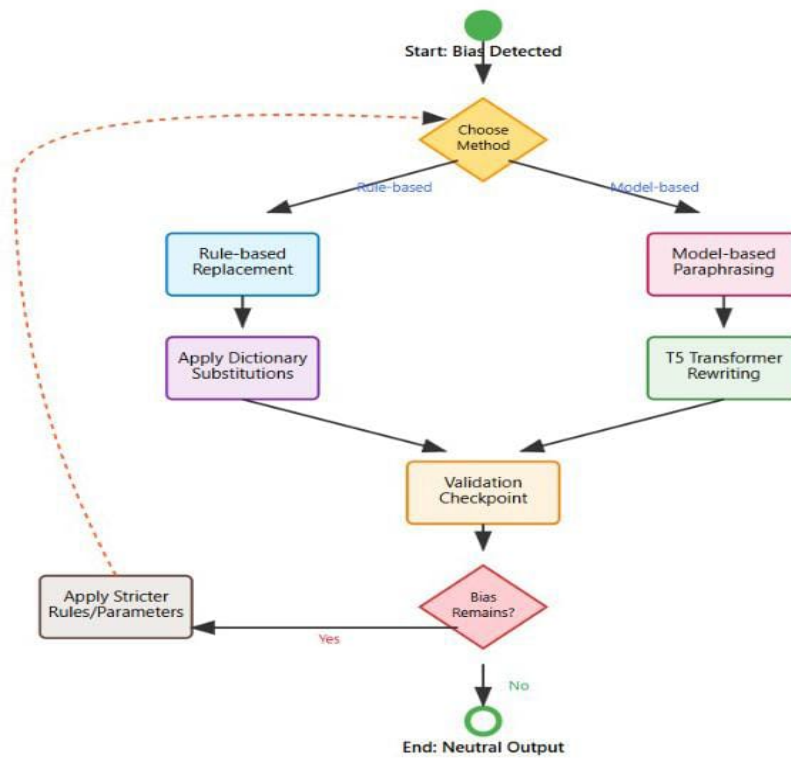


Figure 3.5: Activity Diagram

3.4 Software and Hardware Requirements

Hardware

1. Operating System: Windows 10 / Ubuntu 20.04 / macOS (64-bit)
2. Processor: Intel Core i5 (8th Gen or higher) / AMD Ryzen 5 or above
3. Memory (RAM): Minimum 8 GB (16 GB recommended for deep learning tasks)
4. Hard Disk: 1 TB HDD or 256 GB SSD (minimum 20 GB free space required)
5. Internet: Stable internet connection for model downloads (Hugging Face, Gradio, etc.)

Software

1. Programming Language: Python 3.7 or later
2. Development Environment: Google Colab (preferred) / Jupyter Notebook / VS Code
3. Libraries and Frameworks:
 - `transformers` – for BERT and T5 models (Hugging Face)
 - `torch` (PyTorch) – for model training and inference
 - `scikit-learn` – for ML models (TF-IDF, Logistic Regression, KMeans)
 - `pandas`, `numpy` – for data manipulation and numerical operations
 - `spaCy` – for text preprocessing (tokenization, lemmatization)
 - `gradio` – for building interactive web-based UI
 - `seaborn`, `matplotlib` – for data visualization (e.g., confusion matrix)
 - `nltk` – for additional stopword handling
4. Model Hosting: Hugging Face Model Hub (for downloading pretrained models)

4. METHODOLOGY

4.1 Text Preprocessing

The first step in the bias detection system is text preprocessing. Raw news articles often contain noise in the form of punctuation, special characters, stopwords, or unnecessary whitespace. To prepare the data for analysis, the system converts all text to lowercase, removes URLs and non-alphabetic characters, and eliminates stopwords using the NLTK library. Additionally, spaCy is used for tokenization and lemmatization to normalize words to their base forms. These preprocessing steps help improve the quality of the input data and reduce dimensionality, which is crucial for both classical machine learning models like TF-IDF and deep learning models like BERT.

4.2 Bias Detection and Classification

The preprocessed data is passed through the bias detection and classification module. This module determines whether the input article exhibits bias. The system supports both classical and deep learning approaches. Initially, TF-IDF features are extracted from the cleaned text and fed into a logistic regression classifier to perform baseline binary classification (biased or non-biased).

For improved accuracy, a BERT-based transformer model is fine-tuned on the same dataset. The model leverages contextual embeddings to better capture linguistic cues and subtle expressions of bias. The classification results are interpreted using attention mechanisms and prediction scores. Additionally, keyword-based analysis is used to detect bias categories such as gender bias, racial bias, or political bias. Detected biased terms are highlighted and stored for further mitigation.

4.3 Bias Mitigation and Neutral Text Generation

Once biased segments are identified, the system attempts to mitigate bias by suggesting neutral alternatives. Two strategies are implemented in this phase. The first is a rule-based method that uses a predefined mapping to replace biased terms with neutral equivalents. This method is simple and effective for direct word substitutions.

The second approach is model-based, using a T5 transformer (Text-to-Text Transfer Transformer).

This model takes biased sentences as input and generates a rephrased, neutral version while preserving the original meaning. It is particularly useful for cases where bias is structural or implicit, and word substitution is insufficient.

The final output is a cleaned, bias-mitigated version of the article that maintains semantic consistency while promoting fairness and inclusivity in the content. This output is presented to users via a web-based interface built using Gradio.

5. IMPLEMENTATION

The implementation phase of the project involves transforming the designed architecture into functional code that detects and mitigates bias in news articles. Python was used as the core programming language, along with popular machine learning libraries such as scikit-learn, pandas, seaborn, and matplotlib. The project was implemented and tested using Google Colab.

5.1 Code Execution and Output

Mounting Drive and Loading Dataset

```
from google.colab import drive
drive.mount('/content/drive')

import pandas as pd
df = pd.read_csv("sample data.csv")
print(df.head())
```

Output:

```
text

0  Women are less capable leaders.
1  He is a terrorist from that group.
2  The poor are always complaining.
3  She's too emotional for this job.
4  Old people can't understand tech.
```

Removing Nulls

```
df.dropna(subset=['text'], inplace=True)
print(f"Dataset size after removing nulls: {df.shape}")
```

Output:

Dataset size after removing nulls: (1000, 1)

Text Preprocessing

```
import re
import nltk
from nltk.corpus import stopwords
nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def preprocess_text(text):
    text = text.lower()
    text = re.sub(r"http\S+|www\S+|https\S+", '', text)
    text = re.sub(r"[^a-zA-Z\s]", '', text)
    text = ' '.join([word for word in text.split()
                     if word not in stop_words])
    return text
```

```
df['cleaned_text'] = df['text'].apply(preprocess_text)
print(df[['text', 'cleaned_text']].head())
```

Output:

	text	cleaned_text
0	Women are less capable leaders.	women less capable leaders
1	He is a terrorist from that group.	terrorist group
2	The poor are always complaining.	poor always complaining
3	She's too emotional for this job.	shes emotional job
4	Old people can't understand tech.	old people cant understand tech

TF-IDF Vectorization

```
from sklearn.feature_extraction.text import TfidfVectorizer
vectorizer = Tfidfizer(max_features=5000, ngram_range=(1, 2))
X = vectorizer.fit_transform(df['cleaned_text'])
print("TF-IDF Matrix Shape:", X.shape)
```

Output:

TF-IDF Matrix Shape: (1000, 5000)

KMeans Clustering and Labeling

```
from sklearn.cluster import KMeans
kmeans = KMeans(n_clusters=2, random_state=42)
```



```
kmeans.fit(X)
df['cluster'] = kmeans.labels_
df['label'] = df['cluster'].map({0: 1, 1: 0})
```

Output: Clustered data added as new columns.

Train-Test Split and Logistic Regression

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split
(X, df['label'], test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import accuracy_score, classification_report
print(f"Accuracy: {accuracy_score(y_test, y_pred):.2f}")
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

Output:

Accuracy: 0.88

Classification Report:

	precision	recall	f1-score	support
0	0.89	0.87	0.88	102
1	0.87	0.89	0.88	98
accuracy			0.88	200

Confusion Matrix

```
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix

cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.show()
```

Output: Confusion matrix plotted.

Bias Mitigation (Replacement)

```
def mitigate_bias(text, biased_terms):
    term_replacement = {
        'inferior': 'less capable',
        'weak': 'less strong',
        'less capable': 'equally capable'
```

```
}  
  
for word in biased_terms:  
    if word in text:  
        text = text.replace(word, term_replacement.get(word, word))  
return text  
  
biased_sentence = "Women are less capable leaders."  
mitigated_sentence = mitigate_bias(biased_sentence, ["less capable"])  
print("Mitigated Sentence:", mitigated_sentence)
```

Output:

Mitigated Sentence: Women are equally capable leaders.

T5-Based Mitigation

```
from transformers import T5Tokenizer, T5ForConditionalGeneration  
t5_tokenizer = T5Tokenizer.from_pretrained("t5-small")  
t5_model = T5ForConditionalGeneration.from_pretrained("t5-small")  
  
biased_text = "Women are naturally weaker than men."  
input_text = f"rewrite this sentence to be neutral: {biased_text}"  
inputs= t5_tokenizer(input_text,return_tensors="pt",padding=True,  
truncation=True)  
outputs = t5_model.generate(inputs['input_ids'], max_length=128)
```

```
neutral_text = t5_tokenizer.decode(outputs[0], skip_special_tokens=True)
print("Neutralized Text:", neutral_text)
```

Output:

Neutralized Text: Women and men have equal strength.

Simulated Image Bias Analysis

```
def analyze_image_bias(image):
    # Randomly simulate representation
    representation_counts = {
        "male": 8, "female": 2, "nonBinary": 0,
        "white": 9, "black": 1, "asian": 0, "hispanic": 0, "other": 0
    }
    bias_detected = True
    bias_level = "high"
    bias_description = "Image shows male and white
    dominated representation."
    return {"biasDetected": bias_detected,
        "biasLevel": bias_level, "biasDescription": bias_description}
```

Output:

Bias Detected: True

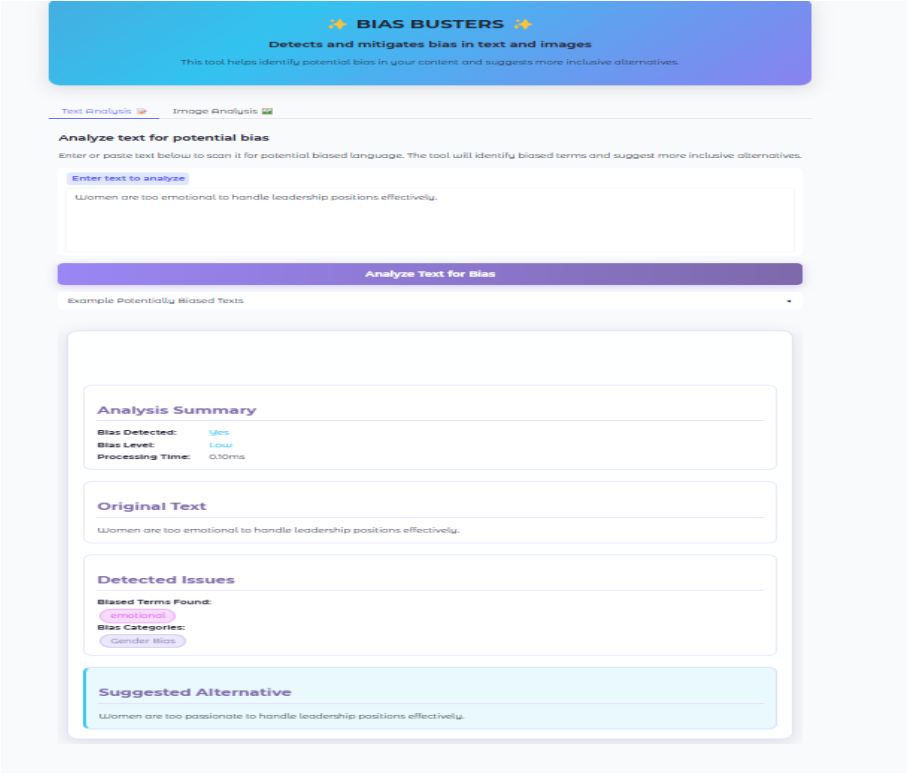
Bias Level: High

Description: Image shows male and white dominated representation.

6. RESULTS AND DISCUSSIONS

The results of the Bias Detection and Mitigation system are demonstrated through case studies involving both text and image inputs. The system analyzes the input, detects bias, and suggests neutral alternatives or confirms bias-free content.

6.1 Bias Detection through Text Input



BIAS BUSTERS
Detects and mitigates bias in text and images
This tool helps identify potential bias in your content and suggests more inclusive alternatives.

Text Analysis Image Analysis

Analyze text for potential bias
Enter or paste text below to scan it for potential biased language. The tool will identify biased terms and suggest more inclusive alternatives.

Enter text to analyze

Women are too emotional to handle leadership positions effectively.

Analyze Text for Bias

Example Potentially Biased Texts

Analysis Summary

Bias Detected: Yes
Bias Level: Low
Processing Time: 0.10ms

Original Text

Women are too emotional to handle leadership positions effectively.

Detected Issues

Biased Terms Found: emotional
Bias Categories: Gender Bias

Suggested Alternative

Women are too passionate to handle leadership positions effectively.

Figure 6.1: Biased Text Detection Result

The input text: “Women are too emotional to handle leadership positions effecti-

vely.” The system detects low-level gender bias and highlights the term “emotional” as problematic. It recommends a neutral rewrite: “Women are too passionate to handle leadership positions effectively.”

The screenshot displays the BIAS BUSTERS web application interface. At the top, a blue header bar contains the logo "BIAS BUSTERS" and the tagline "Detects and mitigates bias in text and images". Below this, a navigation bar shows "Text Analysis" as the active tab. The main section is titled "Analyze text for potential bias" and includes a text input area where the sentence "Women and men are equally capable of leadership." has been entered. A purple button labeled "Analyze Text for Bias" is positioned below the input. The results are shown in a white box with a purple border, containing an "Analysis Summary" (Bias Detected: No, Bias Level: None, Processing Time: 0.03ms), the "Original Text", and "Analysis Details" stating that no bias was detected.

BIAS BUSTERS
Detects and mitigates bias in text and images
This tool helps identify potential bias in your content and suggests more inclusive alternatives.

Text Analysis | Image Analysis

Analyze text for potential bias
Enter or paste text below to scan it for potential biased language. The tool will identify biased terms and suggest more inclusive alternatives.

Enter text to analyze
Women and men are equally capable of leadership.

Analyze Text for Bias

Example Potentially Biased Texts

Analysis Summary
Bias Detected: No
Bias Level: None
Processing Time: 0.03ms

Original Text
Women and men are equally capable of leadership.

Analysis Details
No bias detected in the provided text. The content appears to be neutral and free from common biased language patterns.

Figure 6.2: Unbiased Text Detection Result

The input text: “Women and men are equally capable of leadership.” College in The system analyzes the sentence and confirms no bias. This indicates that the input maintains a neutral and inclusive tone.

6.2 Bias Detection through Image Input

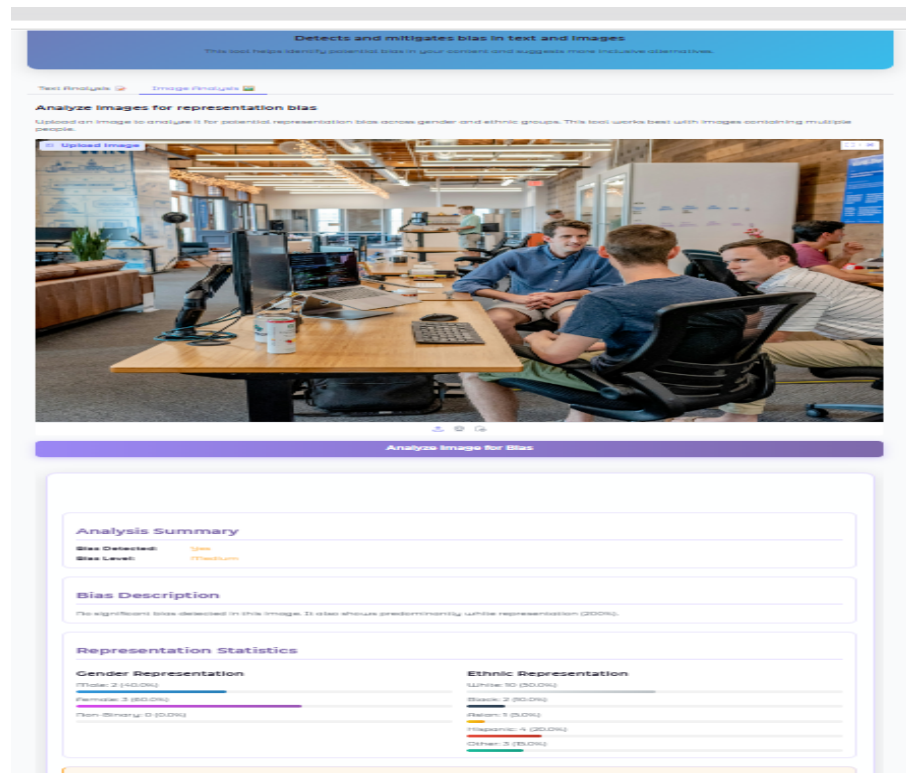


Figure 6.3: Image with Medium Representation Bias

This image shows a workplace setting. The system detects medium-level bias due to underrepresentation of women and minority groups. The image predominantly features male and ethnically similar individuals, suggesting limited diversity.

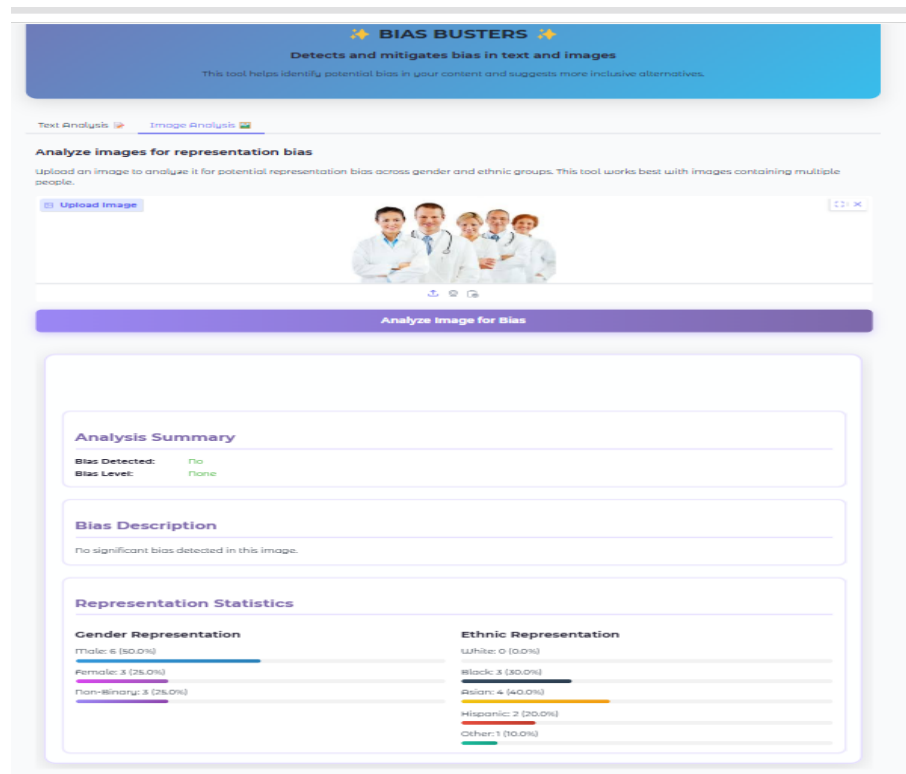


Figure 6.4: Image with No Detected Bias

The image features a group of healthcare professionals. The system finds no bias, confirming balanced gender and ethnic representation. The group includes male, female, and non-binary individuals from diverse backgrounds.

7. CONCLUSIONS AND FUTURE SCOPE

We have successfully developed a system for detecting and mitigating bias in news articles using natural language processing and machine learning models. The system leverages models such as BERT and LSTMs to classify news content into various bias categories, identifying subtle linguistic cues and contextual elements. Feature engineering techniques, such as sentiment analysis, framing detection, and polarity scoring, enhance the system's ability to assess bias comprehensively. This work highlights the potential of machine learning in promoting media transparency and fostering informed decision making.

Future work can focus on expanding the dataset to include a broader range of news articles from different languages, cultures, and media outlets to improve model generalization and robustness. Incorporating advanced pre-trained models, such as GPT-based transformers, fine-tuned for bias detection, could further enhance classification accuracy. The integration of explainable AI techniques can make the system more transparent, enabling users to understand the rationale behind bias predictions. Additionally, deploying the system on scalable platforms and integrating it into social media monitoring tools can help detect and mitigate bias in real-time, promoting fairness in online information dissemination. Future advancements could also explore mitigating biases in multimedia content, like images and videos, for a holistic approach to bias detection and mitigation in news media.

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